

Bayesian Optimization with Fisher Information Geometry: Gradient Bounds and Trust-Region Methods

Saksham Kiroriwal¹, Julius Pfrommer¹, and Jürgen Beyerer^{1,2}

¹Cognitive Industrial Systems, Fraunhofer IOSB, Karlsruhe, Germany

²Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany

{saksham.kiroriwal, julius.pfrommer, juergen.beyerer}@iosb.fraunhofer.de

Abstract

We study Bayesian optimization (BO) through the lens of information geometry. Pulling back the Fisher information metric through the surrogate posterior map yields a local sensitivity tensor on the input space, which leads to an upper bound on the gradient of reparameterizable acquisition functions. This view explains vanishing-gradient behavior in high-dimensional BO and provides a common interpretation of heuristics such as RAASP and dimension-scaled lengthscales. Building on this analysis, we propose FITR, a trust-region-based BO method that replaces lengthscale-based scaling by local pullback-Fisher weights. FITR is not restricted to GP kernels with explicit lengthscales. On GP benchmarks with an SE kernel, experiments show competitive performance using FITR. The proposed method also easily generalizes to non-isotropic surrogates although the gains are more task-dependent in that setting.

1 Introduction

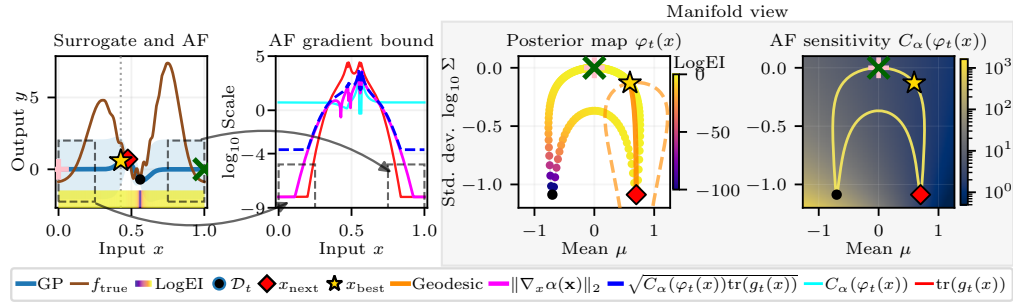


Figure 1: First: Conventional BO view with a GP surrogate (SE kernel) and LogEI [1] acquisition function (AF). Second: Visualization of Proposition 4.2, where the acquisition gradient is upper-bounded by the product of the trace of the pullback Fisher tensor and the sensitivity factor of the AF. Third: Induced geometry on the input space using the pullback Fisher tensor. Fourth: Sensitivity C_α (Section 4) of the AF with respect to the predictive-distribution parameters on the statistical manifold. $\star$ denotes $x = 0$ and $\times$ denotes $x = 1$ in input space (the same markers are used to denote the respective points in the manifold view). Gray boxes indicate regions with no observations where the trace of the pullback Fisher tensor collapses.

Bayesian Optimization (BO) is a popular framework for optimizing black-box functions with expensive evaluations [2]. It has proven to be effective in hyperparameter tuning [3], robotics policy

search [4], and engineering design [5]. As explained in Section 2, BO alternates between fitting a probabilistic surrogate $\mathcal{M}_t$ to the observed data $\mathcal{D}_t$ and optimizing an acquisition function (AF) $\alpha(\cdot)$ to select the next input candidate $\mathbf{x}_{t+1}$. In practice, the inner AF optimization is often non-trivial. The AF is usually non-convex, depends on the surrogate through a predictive distribution, and becomes increasingly difficult to optimize as the input dimension D_x grows.

This issue is now well documented in the high-dimensional BO (HDBO) literature. Recent work shows that both surrogate training and AF optimization can suffer from vanishing gradients in large dimensions [6–8]. Proposed fixes include dimensionally scaled lengthscales priors [6, 7], Random Axis Aligned Subspace Perturbations (RAASP) initializations [9], conformal mappings [10], and trust-region-based search [11]. A central question remains: what common mechanism makes AF optimization collapse, and how much of that mechanism is tied to the surrogate model?

Inspired by the work of Herrmann et al. [12], we analyze BO from a different perspective. As shown in Figure 1, instead of looking at the AF only as a function over $\mathcal{X}$, we utilize the chain rule and information geometry. The surrogate $\mathcal{M}_t$ is a map from input space $\mathcal{X}$ to the statistical manifold $\mathcal{S}$ of predictive distributions. These predictive distribution parameters, like mean and variance, are the parameters that the AF really depends on. Information geometry [13, 14] becomes a natural way to study how the surrogate induces geometry on the input space through the pullback Fisher tensor. This induced geometry tells us how strongly local perturbations in $\mathbf{x}$ change the predictive distribution.

This view separates two effects that are usually entangled in AF optimization. The acquisition design determines how sensitive the utility is to changes in the predictive distribution. The surrogate-induced geometry determines whether moving in input space produces such changes at all. Section 4 formalizes this separation through an acquisition-gradient bound. This perspective then becomes a common analytical tool for understanding HDBO:

- In Section 4, we analyze the pullback geometry induced by the surrogate posterior map and derive a bound for the AF gradient norm $\|\nabla_{\mathbf{x}}\alpha\|_2$ that separates acquisition-side sensitivity from surrogate-induced geometry.
- We use this factorization to analyze vanishing-gradient behavior in HDBO under a common framework. This yields a unified interpretation of dimensionally scaled priors [6, 7] and RAASP [8, 9].
- In Section 5, we propose the Fisher-Information Trust Region (FITR), which replaces kernel-lengthscale scaling by local pullback-Fisher weights. FITR applies to differentiable probabilistic surrogates that satisfy the stated regularity conditions and is not restricted to GP kernels with explicit lengthscales. Section 6 shows that FITR improves on TuRBO-REI on several benchmarks and remains competitive with strong baselines on the remainder.

2 Preliminaries

The aim of BO is to maximize a black-box function f_{true} over a feasible set $\mathcal{X} \subset \mathbb{R}^{D_x}$. At each BO iteration t , we assume that we have observed data $\mathcal{D}_t = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$, where input $\mathbf{x}_n \in \mathcal{X}$ and output $y_n = f_{\text{true}}(\mathbf{x}_n) + \varepsilon_n$. We allow $y_n \in \mathbb{R}^{D_y}$ to be multi-dimensional and assume $\varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2 I_{D_y})$, which reduces to $\mathcal{N}(0, \sigma_\varepsilon^2)$ in the scalar-output case. BO alternates between fitting a probabilistic surrogate $\mathcal{M}_t$ and maximizing an AF via $\arg \max_{\mathbf{x} \in \mathcal{X}} \alpha(\mathbf{x}; \mathcal{D}_t)$ to obtain the next input candidate $\mathbf{x}_{t+1}$ [15, 16]. We then evaluate $y_{t+1} = f_{\text{true}}(\mathbf{x}_{t+1})$ and update the observed data $\mathcal{D}_{t+1} = \mathcal{D}_t \cup \{(\mathbf{x}_{t+1}, y_{t+1})\}$. We repeat until a stopping criterion is met.

Surrogate Models At each iteration t , the surrogate model $\mathcal{M}_t$ models the predictive distribution $p(y|\mathbf{x}, \mathcal{D}_t) = p(y; \boldsymbol{\theta}_{t,\mathbf{x}})$, where $\boldsymbol{\theta}_{t,\mathbf{x}} \in \mathcal{S}$ are the parameters of the predictive distribution [17]. The hyperparameters $\boldsymbol{\lambda}_t$ of $\mathcal{M}_t$ specify the distribution parameters $\boldsymbol{\theta}_{t,\mathbf{x}} \leftarrow \mathcal{M}_t(\mathbf{x}; \mathcal{D}_t)$ and can be trained by maximizing the marginal log-likelihood [18] or another suitable objective [19, 20].

As shown in the manifold view in Figure 1, the surrogate model can also be interpreted as a smooth map from inputs to a family of probability distributions. Each input $\mathbf{x} \in \mathcal{X}$ is mapped to a point $\boldsymbol{\theta}_{t,\mathbf{x}}$ on the statistical manifold $\mathcal{S}$, the set of all distributions the surrogate can produce [13]. Input locations that appear far apart in Euclidean space may be close in the manifold view because the surrogate induces its own local geometry, as shown by the $+$ and $\times$ markers in Figure 1. This map is at least C^1 differentiable, a mild condition satisfied by many popular surrogate models such as

Gaussian processes (GPs) with differentiable kernels [18] or Bayesian neural networks (BNNs) with smooth activations [21]. The *posterior map* φ_t induced by $\mathcal{M}_t$ can be defined as

$$\varphi_t : \mathcal{X} \rightarrow \mathcal{S}, \quad \mathbf{x} \mapsto \boldsymbol{\theta}_{t,\mathbf{x}}, \text{ such that } \boldsymbol{\theta}_{t,\mathbf{x}} \in \mathcal{S}, \quad (1)$$

A common choice for the surrogate model is a Gaussian process (GP). For a scalar output $y \in \mathbb{R}$ at an input location $\mathbf{x} \in \mathcal{X}$, $\mathcal{M}_t$ models $\boldsymbol{\theta}_{t,\mathbf{x}} = (\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$ and gives $p(y|\mathbf{x}, \mathcal{D}_t) = \mathcal{N}(y; \mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$ with $\mathcal{S} = \mathbb{R} \times \mathbb{R}_+$ [18]. GPs are discussed in further detail in Appendix A.

Acquisition Function (AF) The AF α is the expected utility over the predictive posterior evaluated at a candidate location $\mathbf{x} \in \mathcal{X}$ and can be defined as $\alpha(\mathbf{x}) = \mathbb{E}_y [a(y; \boldsymbol{\theta}_{t,\mathbf{x}}, \boldsymbol{\xi})]$, where a is the associated utility function for the AF and $\boldsymbol{\xi}$ are respective acquisition parameters. The next query is $\mathbf{x}_{t+1} = \arg \max_{\mathbf{x}} \alpha(\mathbf{x}; \mathcal{D}_t)$ [16, 17, 22]. Since $p(y|\mathbf{x}, \mathcal{D}_t)$ can be parameterized by $\boldsymbol{\theta}_{t,\mathbf{x}}$, α depends on $\mathbf{x}$ only through $\boldsymbol{\theta}_{t,\mathbf{x}}$. Using the reparameterization trick [23], we introduce a common notation to represent both closed-form and Monte Carlo AFs as

$$\alpha(\mathbf{x}) = \mathbb{E}_{\boldsymbol{\zeta}} [\bar{\alpha}(\boldsymbol{\theta}_{t,\mathbf{x}}, \boldsymbol{\zeta}; \boldsymbol{\xi})], \quad (2)$$

where $\boldsymbol{\zeta}$ is base noise drawn independently of $\mathbf{x}$ [17, 23]. However for closed-form AFs, $\bar{\alpha}$ is independent of $\boldsymbol{\zeta}$ and (2) can be trivially written as $\alpha(\mathbf{x}) = \bar{\alpha}(\boldsymbol{\theta}_{t,\mathbf{x}}; \boldsymbol{\xi})$. Due to this, the closed-form AF and the $\boldsymbol{\zeta}$ -dependent integrand used in the MC form can, in general, be different formulations of $\boldsymbol{\theta}_{t,\mathbf{x}}$. Appendix B discusses this in detail and lists both forms for different AFs.

Fisher Information Matrix For a parametric distribution family $p(y; \boldsymbol{\theta})$ with $\boldsymbol{\theta} \in \mathcal{S}$, the *Fisher information matrix* (FIM) $\mathcal{I}_{\mathcal{S}}(\boldsymbol{\theta}) \in \mathbb{R}^{D_{\boldsymbol{\theta}} \times D_{\boldsymbol{\theta}}}$ is the expected outer product of $\nabla_{\boldsymbol{\theta}} \log p(y; \boldsymbol{\theta})$ as

$$[\mathcal{I}_{\mathcal{S}}(\boldsymbol{\theta})]_{ij} = \mathbb{E}_{p(y; \boldsymbol{\theta})} \left[\frac{\partial \log p(y; \boldsymbol{\theta})}{\partial \theta_i} \frac{\partial \log p(y; \boldsymbol{\theta})}{\partial \theta_j} \right]. \quad (3)$$

Amari and Nagaoka [24] showed that the FIM induces a Riemannian metric, known as *Fisher–Rao metric*, on the statistical manifold $\mathcal{S}$. The Fisher–Rao metric explains the local geometry of the manifold [13, 25]. It quantifies the sensitivity of the predictive distribution $p(y; \boldsymbol{\theta})$ to changes in $\boldsymbol{\theta}$.

Riemannian Geometry A *Riemannian manifold* $(\mathcal{M}, g)$ is a smooth manifold $\mathcal{M}$ equipped with a Riemannian metric g , whose pointwise evaluation g_p defines a positive definite inner product on each tangent space $T_p \mathcal{M} \simeq \mathbb{R}^{d_p}$ [14]. The geodesic distance between $p_1, p_2 \in \mathcal{M}$ is the minimum path length on the manifold that connects the two points and is given by

$$d_g(p_1, p_2) = \inf_{\gamma} \int_0^1 \sqrt{g_{\gamma(t)}(\dot{\gamma}(t), \dot{\gamma}(t))} dt, \quad (4)$$

where the infimum is over smooth curves $\gamma : [0, 1] \rightarrow \mathcal{M}$ with $\gamma(0) = p_1$ and $\gamma(1) = p_2$ [14]. The Fisher–Rao metric $\mathcal{I}_{\mathcal{S}}(\boldsymbol{\theta})$ induces a Riemannian structure on the statistical manifold $\mathcal{S}$ [13]. We define the pullback Fisher tensor through the φ_t and discuss its properties in Section 4.

3 Related Work

Information geometry and applications. Natural gradient descent (NGD) is a key application of this theory [25, 26]. Work by Khan and Rue [27] shows how algorithms such as stochastic gradient descent (SGD) and Kalman filters can be derived from an algorithm called *Bayesian Learning Rule*, and how natural gradients can improve the resulting updates. As another application of information geometry, Kim et al. [28] presented Fisher–Sharpness–Aware Minimization (FSAM), which replaced the Euclidean ball in sharpness-aware minimization with a Fisher ellipsoid, improving generalization and robustness. Beik-Mohammadi et al. [29] learn Riemannian manifolds for geodesic motion generation in robotics.

Geometry-aware Bayesian optimization. Jaquier et al. [4], Jaquier and Rozo [30] introduced geometry-aware kernels that used geodesic distances in Bayesian optimization when the search space is non-Euclidean, such as robot orientation. However, in our work, we explore the geometry that is induced in the search space by the posterior map of the surrogate model $\mathcal{M}_t$ as mentioned in (1). We do not assume that the search space has any intrinsic geometric structure. Recent work by Yu et al.

[31] proposes GIT-BO with Tabular Foundation Model (TabPFNv2) as the surrogate model. They use the gradient of the predictive mean to identify a global active subspace for high-dimensional BO. Our use of Fisher geometry is local and trust-region based: we shape candidate regions around the current center using the predictive distribution, not only the mean. A direct comparison with GIT-BO would therefore mix surrogate choice with trust-region geometry.

Pullback metrics for probabilistic models. Tosi et al. [32] proposed the idea of pulling back a Riemannian metric for Gaussian Process Latent Variable Models (GPLVMs). They derived the expected metric tensor and used the resulting geodesics for improving data interpolation. Arvanitidis et al. [33] showed how the latent space can be characterized by a stochastic Riemannian metric, resulting in improved sampling and interpolation in latent space. Arvanitidis et al. [34] used the Fisher-Rao metric associated with the space of decoder distributions, enabling meaningful latent geometries for a broader class of decoders. Rozo et al. [35] generalized the construction to wrapped GPLVMs defined on Riemannian data manifolds, where they pull back a Riemannian structure from the data manifold to the latent space.

High-dimensional Bayesian optimization. Recent work by Hvarfner et al. [6], Xu et al. [7] has shown that BO can be effective in high dimensions (HDBO). They analyze the vanishing-gradient problem in surrogate-model and acquisition-function (AF) optimization in high dimensions. They propose a dimensionally scaled lengthscale prior to avoid the vanishing gradient problem. Papenmeier et al. [8] confirm the finding and propose $\sqrt{D_x}/10$ initialization of lengthscales for Maximum Likelihood Estimation (MLE) for GP surrogates. They also show that RAASP for AF optimization helps in preventing vanishing gradients.

Local search for global optimization and trust-region methods offer complementary approaches. TuRBO is a popular trust-region-based method proposed by Eriksson et al. [11] to address high dimensionality. They use local axis-aligned hyperrectangles to define the trust region around the current best point. Work by Papenmeier et al. [9, 36] extends trust-region-based methods to combinatorial and mixed spaces as well. However, existing trust-region constructions in this setting rely on GP surrogates with explicit lengthscale parameters like squared exponential kernel or Matern kernel.

4 Manifold View of Bayesian Optimization

As mentioned in Section 2, the acquisition function depends on $\mathbf{x}$ through the surrogate’s predictive parameters $\theta_{t,\mathbf{x}}$. Viewing the surrogate as a smooth map $\varphi_t : \mathcal{X} \rightarrow \mathcal{S}$ into the statistical manifold of predictive distributions shifts the analysis of acquisition-gradient behavior to the geometry of this map. Pulling back the Fisher–Rao metric from $\mathcal{S}$ to $\mathcal{X}$ via φ_t yields the pullback Fisher tensor $\mathbf{g}_t(\mathbf{x})$, which quantifies how sensitively $p(y; \theta_{t,\mathbf{x}})$ changes under local perturbations of the input.

We define the pullback Fisher tensor and establish an acquisition gradient bound (Section 4.1). Then in Section 4.2 we show how the critical λ scaling of $\sqrt{D_x}$ from [6–8] and other HDBO findings like RAASP can be viewed from the geometric perspective of the pullback Fisher tensor.

Assumption 4.1 (Surrogate Differentiability and Positive Fisher Metric). *φ_t is C^1 on $\mathcal{X}$ (satisfied by GPs with differentiable kernels and by BNNs with smooth activations) and $\mathcal{I}_{\mathcal{S}}(\varphi_t(\mathbf{x})) \succ 0$ for all $\mathbf{x} \in \mathcal{X}$ (a minimum noise $\sigma_\epsilon^2 > 0$ ensures this for GP regression).*

As discussed in Appendix D.1, the first condition makes φ_t differentiable so that the Jacobian J_t exists. The second ensures $\mathcal{I}_{\mathcal{S}}$ is a valid inner product on $\mathcal{S}$, so its inverse (required in the acquisition sensitivity (7)) is well-defined. The pullback Fisher tensor quantifies the sensitivity of the predictive distribution to perturbations in the input space at each input location.

Definition 4.1 (Pullback Fisher Tensor). *Under Assumption 4.1, the pullback Fisher tensor induced on the input space $\mathcal{X}$ by the predictive distribution manifold $\mathcal{S}$ through the posterior map φ_t is*

$$\mathbf{g}_t(\mathbf{x}) = J_t(\mathbf{x})^\top \mathcal{I}_{\mathcal{S}}(\varphi_t(\mathbf{x})) J_t(\mathbf{x}) \in \mathbb{R}^{D_x \times D_x}, \quad (5)$$

where $\mathbf{J}_t(\mathbf{x}) = \partial \theta_{t,\mathbf{x}} / \partial \mathbf{x} \in \mathbb{R}^{D_\theta \times D_x}$ is the Jacobian of the posterior map.

Figure 1 shows the induced geometry on a 1D Euclidean input space by the GP surrogate with SE kernel and LogEI AF using the pullback metric tensor. Figure 2 similarly shows the trace of the

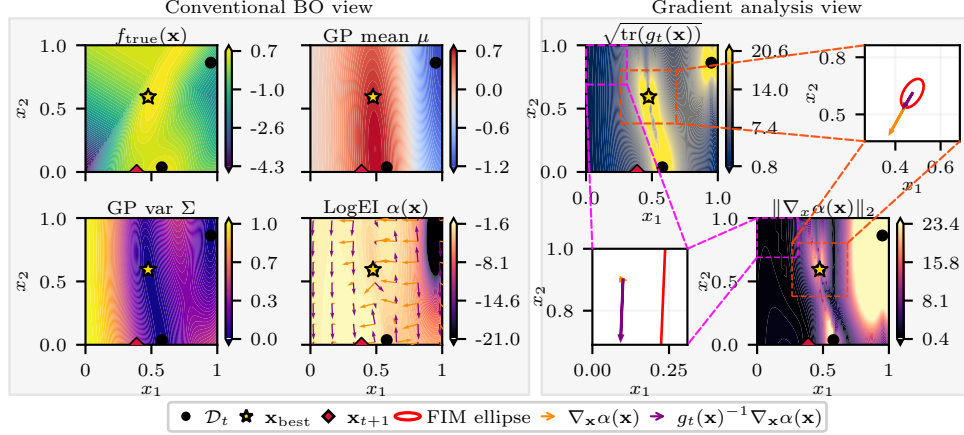


Figure 2: Conventional view of BO with a GP surrogate (SE kernel) and LogEI AF. The LogEI plot also shows the gradient (orange) and natural gradient (purple) vector field in the input space. The "gradient analysis" shows pullback metric trace $\sqrt{\text{tr}(\mathbf{g}_t(\mathbf{x}))}$ and AF gradient norm $\|\nabla_{\mathbf{x}}\alpha(\mathbf{x})\|_2$. Trace is non-degenerate near the observed data and collapses in regions where the GP posterior has reverted to the prior (no observations). The red ellipses are calculated using the pullback metric tensor at respective input locations.

pullback metric tensor for a 2D input space. Since for $D_x > 1$, the pullback metric is a tensor, the eigenvalues of $\mathbf{g}_t(\mathbf{x})$ reflect how strongly the predictive distribution changes with respect to perturbations in input space at $\mathbf{x}$ in respective directions. Large eigenvalues correspond to directions (not necessarily axes-aligned with $\mathcal{X}$) along which the distribution changes rapidly, small or zero eigenvalues to directions that are locally uninformative. This effect can be seen in the "Gradient Analysis" panel of Figure 2, where we show the $\nabla_{\mathbf{x}}\alpha(\mathbf{x})$ and the natural gradient $\mathbf{g}_t(\mathbf{x})^{-1}\nabla_{\mathbf{x}}\alpha(\mathbf{x})$ vector field and also zoomed in view for two input locations denoted by the magenta and red boxes.

The pullback tensor $\mathbf{g}_t(\mathbf{x})$ is a Riemannian metric only when it is positive definite. Since $\mathbf{g}_t$ has rank at most D_θ (at most 2 for a univariate GP), the induced geometry on $\mathcal{X}$ is degenerate when $D_x > D_\theta$. Locally, perturbations at $\mathbf{x}$ in null directions leave the predictive parameters $\theta_{\mathbf{x}}$ unchanged.

4.1 Acquisition Gradient Bound via Pullback Fisher Tensor

Proposition 4.2 (Acquisition Gradient Bound). *Let Assumption 4.1 hold. For any $\mathbf{x} \in \mathcal{X}$, let $\theta_{t,\mathbf{x}} = \varphi_t(\mathbf{x})$ and suppose the acquisition admits the representation $\alpha(\mathbf{x}) = \mathbb{E}_{\zeta}[\bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta)]$, where the interchange of gradient and expectation is valid, for example under the scalar Gaussian reparameterization conditions of Appendix D.1, and $\mathbb{E}_{\zeta}[\|\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}\|^2] < \infty$. Then*

$$\|\nabla_{\mathbf{x}}\alpha(\mathbf{x})\|_2 \leq \sqrt{C_{\alpha}(\theta_{t,\mathbf{x}}) \cdot \text{tr}(\mathbf{g}_t(\mathbf{x}))}, \quad (6)$$

where the acquisition sensitivity is

$$C_{\alpha}(\theta_{t,\mathbf{x}}) = \mathbb{E}_{\zeta} \left[\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta)^{\top} \mathcal{I}_{\mathcal{S}}(\theta_{t,\mathbf{x}})^{-1} \nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta) \right]. \quad (7)$$

Proof is given in Appendix D.2. Equation 6 separates the roles of the surrogate and the acquisition in the inner BO optimization problem. The trace $\text{tr}(\mathbf{g}_t(\mathbf{x}))$ depends only on the surrogate map and measures how strongly the predictive distribution changes under local perturbations at $\mathbf{x}$. Equation 7 defines the *acquisition-side sensitivity* of the AF value to perturbations on the predictive manifold $\mathcal{S}$. For a fixed AF and predictive distribution family, the formula for acquisition-side sensitivity depends on the current predictive state θ and the manifold metric $\mathcal{I}_{\mathcal{S}}(\theta)$, but not on which surrogate produced that state. Surrogates represented through the same Gaussian predictive family, such as GP posteriors and Gaussian predictive approximations to Bayesian neural networks (BNNs), therefore use the same acquisition-side formula when evaluated at the same predictive parameters. The surrogate-specific effect enters through $\text{tr}(\mathbf{g}_t(\mathbf{x}))$. Appendix D.4 discusses C_{α} for common AFs under a Gaussian predictive distribution.

When the surrogate maps a region of input space to a near-constant predictive distribution, or when the posterior reverts to the prior away from the data, $\text{tr}(\mathbf{g}_t(\cdot))$ collapses in that region. Proposition 4.2

then shows that bounded acquisition-side sensitivity forces the acquisition gradient to become small there. This effect is visible in Figure 1 and Figure 2: the AF gradient vanishes where the pullback trace collapses and remains active where the local predictive geometry is non-degenerate.

4.2 Unified FIM-based Framework for HDBO analysis

Lengthscale Scaling and Random Axis Aligned Subspace Perturbations For GP surrogates with AFs such as EI, UCB, or PI, the sensitivity $C_\alpha(\varphi_t(\mathbf{x}))$ remains uniformly bounded as D_x grows, as shown in Appendix D.3. The experiments use LogEI, for which the same local regularity argument applies in the strictly positive-variance regime considered here. As shown in Appendix D.5, for i.i.d. inputs drawn uniformly from $[0, 1]^{D_x}$, the expected Euclidean distance $\mathbb{E}[d] = \Theta(\sqrt{D_x})$, where $d = \|\mathbf{x} - \mathbf{x}'\|$, with exact asymptotic $\mathbb{E}[d] \sim \frac{1}{\sqrt{6}}\sqrt{D_x}$ as $D_x \rightarrow \infty$. For the isotropic SE kernel, $k_\lambda(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\frac{d^2}{2\lambda^2})$, so keeping the exponent non-degenerate requires $\lambda = \Theta(\sqrt{D_x})$. Isotropic Matérn kernels depend on the normalized distance d/λ , which leads to the same scaling. If the lengthscale grows more slowly, the posterior map becomes nearly constant away from the data and $\text{tr}(g_t(\mathbf{x}))$ collapses, which forces the AF gradient to become small by Proposition 4.2. This recovers the scaling requirement identified by Hvarfner et al. [6], Xu et al. [7], Papenmeier et al. [8] from the viewpoint of induced geometry. The same trace-collapse argument also helps explain why RAASP around observed data [9] improves multi-start AF optimization. The pullback Fisher tensor therefore provides a common language for both failure modes and fixes in HDBO.

TuRBO connection Figure 2 shows that the pullback trace and the AF gradient remain non-degenerate near the observed data. TuRBO develops on a similar principle, where it centers a local axis-aligned TR at the incumbent and adapts its size over time using the learned lengthscales [11]. The dimensions with longer lengthscales have a longer TR length. Appendix D.6 shows that the metric tensor at a local optimum $\mathbf{x}_{\text{best}}$ is often dominated by the variance-gradient term, which is inversely proportional to the lengthscales. The ellipsoid made using the pullback metric would be longer in the dimensions with longer lengthscales. This is similar to how TuRBO uses the learned lengthscales to create TR. This motivates the use of pullback-metric-based TR for BO, which we call the Fisher-Information Trust Region (FITR). The FITR is a location-dependent TR that can be extended to any surrogate for which the derivative of the predictive distribution with respect to the input variables can be calculated.

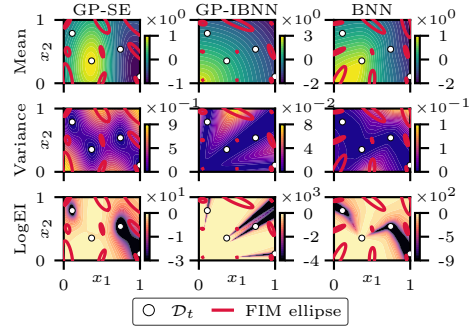


Figure 3: Ellipses using $g(\mathbf{x})$ from (5) for different surrogates: GP with squared exponential (SE) kernel, GP with Infinite Width Bayesian Neural Network (IBNN) kernel [21] and Bayesian Neural Network (BNN).

5 Fisher-Information Trust Region (FITR)

As explained in Appendix B.1, TuRBO centers the trust region at the best observed location $\mathbf{x}_{\text{best}}$ and scales the dimension-wise side lengths using kernel lengthscales [11]. This construction is effective because it avoids performing the inner AF optimization in large regions where the surrogate has little information. Since the kernels are isotropic, the notion of distance underlying TuRBO remains Euclidean. TuRBO has proven effective in practice, but is also limited to SE or Matérn kernels.

Section 4 suggests a different local notion of distance. The acquisition depends on $\mathbf{x}$ through $\varphi_t(\mathbf{x})$, so the relevant local geometry is defined by the pullback Fisher tensor $\mathbf{g}_t(\mathbf{x})$ in (5). This tensor can be computed for any differentiable probabilistic surrogate that satisfies Assumption 4.1. The ideal local Fisher region is the ellipsoid

$$\{\mathbf{x} \in \mathcal{X} : (\mathbf{x} - \mathbf{x}_{t,c})^\top \mathbf{g}_t(\mathbf{x}_{t,c})(\mathbf{x} - \mathbf{x}_{t,c}) \leq \kappa_t^2\}. \quad (8)$$

Limitations. Appendix D.7 shows that the FIM is the second-order Taylor expansion of Kullback-Leibler (KL) divergence [13, 24]. A purely KL-derived trust region is inherently mode-seeking

and can become too exploitative near local optima [37]. We therefore retain TuRBO’s outer trust-region scaling rule, Algorithm 3, to preserve its exploration behavior. A second issue is scalability with D_x , since the full local ellipsoid requires inversion or whitening of the pullback tensor. For scalar predictive families, the pullback rank is at most two, so the tensor is rank-deficient when $D_x > 2$. Without regularization, near-null directions would produce arbitrarily large radii. Popular BO frameworks such as BoTorch also largely support sampling from axis-aligned trust regions. One can create a hyperrectangle from the pullback tensor by using the infinity norm and whitening to make it axis-aligned. However, once such a region intersects the constraints/bounds of $\mathcal{X}$, the feasible set becomes a clipped polytope rather than a simple box. Polytope sampling is substantially slower and less efficient than simple axis-aligned Sobol sampling for multi-start AF optimization.

Proposed method. The experiments in this paper use a diagonal approximation of the pullback Fisher tensor rather than the full ellipsoid in (8). At the trust-region center $\mathbf{x}_{t,c}$, we first estimate the diagonal entries of the unregularized pullback Fisher tensor by Monte Carlo (MC) samples from the predictive posterior:

$$\hat{g}_{t,j}(\mathbf{x}_{t,c}) = \frac{1}{S} \sum_{s=1}^S \left(\frac{\partial}{\partial x_j} \log p(\tilde{y}_s; \varphi_t(\mathbf{x})) \Big|_{\mathbf{x}=\mathbf{x}_{t,c}} \right)^2, \quad \tilde{y}_s \sim p(y; \varphi_t(\mathbf{x}_{t,c})), \quad (9)$$

where S is the number of MC samples and $j \in \{1, \dots, D_x\}$. Appendix D.8 shows that Equation 9 is an unbiased MC estimator of the corresponding diagonal entry of the pullback Fisher tensor [25]. Regularization is added only after this estimation step.

We use diagonal Tikhonov-style regularization [25] to avoid unbounded stretching in directions associated with zero or near-zero eigenvalues of the pullback tensor:

$$\bar{w}_{t,j} = (\hat{\mathbf{g}}_{t,j}(\mathbf{x}_{t,c}) + \varepsilon_t)^{-1/2}.$$

The regularization parameter is chosen adaptively from the local Fisher scale as $\varepsilon_t = \frac{1}{D_x} \sum_{j=1}^{D_x} \hat{\mathbf{g}}_{t,j}(\mathbf{x}_{t,c}) + \varepsilon_{\text{jit}}$, where ε_{jit} is a small numerical jitter to prevent division by zero. This damping adds an isotropic curvature floor and interpolates between induced geometry and Euclidean geometry when the pullback tensor is degenerate [25]. The ℓ^2 norm gives an ellipsoid from (8). When we use the ℓ_∞ norm, we obtain a box-shaped TR. We implement axis-aligned FITR using only the diagonal of the regularized pullback tensor. This yields a location-dependent axis-aligned box that remains compatible with standard BO toolchains and avoids the clipped-polytope sampling problem at the domain boundary. The diagonal approximation also keeps the method practical in higher dimensions compared with the full whitening-based variant. TuRBO computes trust-region weights as in Algorithm 2 and then uses volume normalization and trust-region-bounded AF optimization to select the next candidate, as outlined in Algorithm 3. FITR replaces only the local weight construction in Algorithm 2 and keeps the same volume normalization and outer control logic. The resulting trust region is a box in (10).

$$\mathcal{X}_t^{\text{TR}} = \left\{ \mathbf{x} \in \mathcal{X} : \|\text{diag}(\bar{\mathbf{w}}_t)^{-1}(\mathbf{x} - \mathbf{x}_{t,c})\|_\infty \leq \frac{\kappa_t}{2} \right\}, \quad (10)$$

where $\bar{\mathbf{w}}_t$ denotes the volume-normalized weights obtained from the raw values returned by Algorithm 1. The only algorithmic change relative to TuRBO is therefore the local weight construction: ARD lengthscales are replaced by regularized pullback-Fisher sensitivities, while the trust-region scaling rule and outer control logic remain unchanged. We refer to the full whitening-based construction as FullFITR and treat it as an ablation rather than the default method.

6 Experiments

Problems and baselines. We evaluate FITR with a GP using an SE kernel against strong baselines such as TuRBO [11], DSP [6], and BOUNCE [36] on nine continuous benchmark problems, as

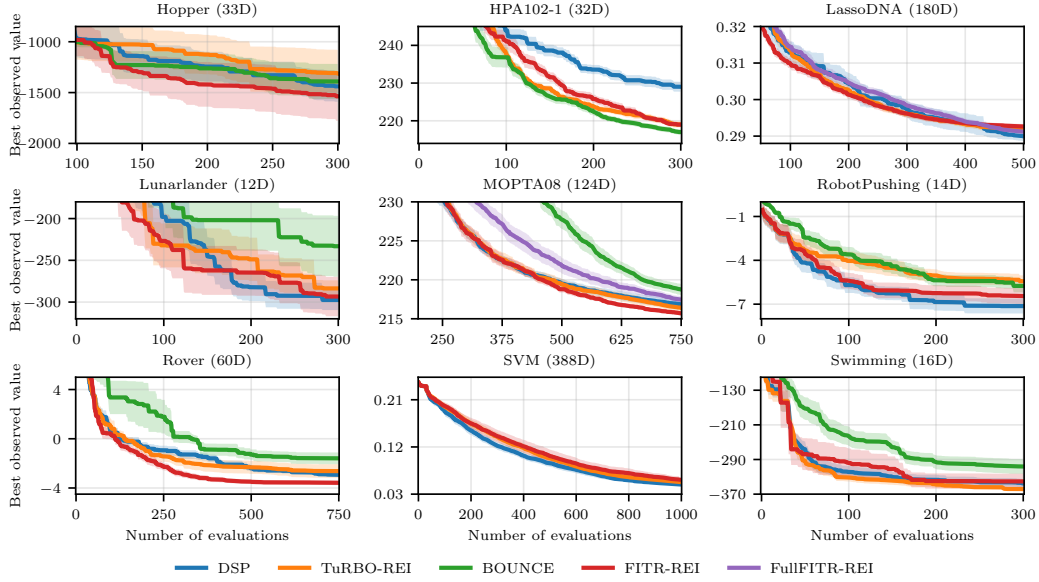


Figure 4: BO performance (lower is better) comparison for different methods with a GP using an SE kernel. Solid curves show means over 11 trials, and shaded bands show standard error. FITR-REI and TuRBO-REI use the same GP modeling pipeline and regional trust-region center selection; they differ only in whether the trust-region shape is determined by ARD lengthscales or by the pullback Fisher tensor. FullFITR-REI was evaluated only on a subset of problems and is discussed separately in the text.

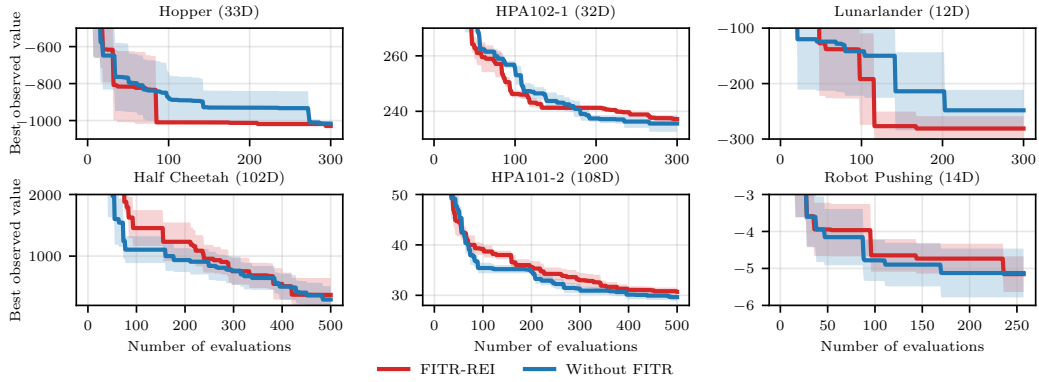


Figure 5: BO performance comparison (lower is better) for different methods with a GP using an IBNN kernel of depth 3. Solid curves show means over 5 trials, and shaded bands show standard error.

shown in Figure 4. These are control and robotics benchmarks commonly used in high-dimensional BO [11, 36]. HPA102-1 is a human-powered aircraft design task from the HPA benchmark suite [38]. MOPTA08 is an automotive design benchmark [11]; LassoDNA and SVM test sparse regression and hyperparameter tuning behavior. Implementation details are given in Appendix C. We also evaluate the same FITR construction with a non-lengthscale kernel, namely IBNN [21], as a transferability check beyond standard ARD kernels. Figures 4 and 5 report the best observed objective value in the benchmark convention, while the BO procedure itself is written in maximization form.

Results Figure 4 shows that, for a GP with an SE kernel, FITR improves on TuRBO-REI on several benchmarks and remains competitive with strong baselines such as DSP and BOUNCE on the remainder. This matches the motivation from Section 5: local Fisher information can provide a useful

location-dependent trust-region shape beyond fitted kernel lengthscales. Figure 6 in Appendix C reports the corresponding trust-region diagnostics. With diagonal Tikhonov regularization, the anisotropy of FITR remains bounded across the benchmark suite while still adapting to local predictive geometry. The trust-region length κ_t for FITR and TuRBO stays qualitatively similar across problems, which suggests that the performance differences in Figure 4 come from the trust-region shape rather than from a different restart or length-adaptation schedule. Figure 6 also shows that the per-iteration candidate-generation time for FITR remains in the same order as TuRBO, with a modest overhead from computing the local Fisher weights.

On the subset of problems where we evaluated FullFITR, it did not outperform diagonal FITR. A likely reason is that the pullback geometry is local. If the geometry varies rapidly near the trust-region center, the rotated region estimated at the center need not model nearby candidate locations better than the diagonal approximation. Boundary clipping and the higher numerical cost of the full tensor may also contribute. We therefore use diagonal FITR as the practical default.

The IBNN experiments in Figure 5 use the same pipeline with a depth-3 IBNN kernel [21]. In this setting we compare FITR only against the non-trust-region baseline, so the figure should be read as a transferability check rather than as a full non-lengthscale benchmark comparison. FITR improves on the non-trust-region baseline on a subset of tasks and remains comparable on the remainder. These results show that the same FITR pipeline can be paired with a non-lengthscale kernel, although the gains for anisotropic kernels in trust-region methods appear task-dependent.

7 Limitations

Proposition 4.2 bounds the acquisition gradient during inner optimization; it does not imply a regret guarantee or a global convergence statement for the outer BO loop. One could also replace the trust region by a KL-divergence constraint and solve the resulting problem with an Augmented Lagrangian Method (ALM) [39]. During our experiments, however, this ALM-based variant became highly exploitative and less exploratory, which degraded performance.

For non-isotropic kernels like IBNNs, the pullback trace need not collapse with D_x in the same way as for many lengthscales-dominated GPs [21], so the vanishing-gradient narrative from Section 4.2 applies less sharply. The improvement in performance using FITR becomes task-dependent. A precise analysis of when local Fisher geometry predicts a measurable search advantage beyond isotropic or deep-kernel setups is left for future work.

8 Conclusion

We propose an information-geometric perspective on Bayesian optimization (BO). The surrogate is treated as a smooth map from the input space to the manifold of predictive distributions, and reparameterizable acquisition functions depend on the input only through these predictive parameters. Proposition 4.2 separates the norm of the acquisition gradient into two factors: an acquisition-side sensitivity and the trace of the pullback Fisher tensor induced by the surrogate. For bounded acquisition-side sensitivity, the acquisition gradient is vanished when the pullback trace collapses. This provides a common explanation for several empirical observations in high-dimensional BO, including dimension-scaled lengthscales and local initialization around observed data.

Using these insights, we propose the Fisher-Information Trust Region (FITR), a simple trust-region method for differentiable probabilistic surrogates that satisfy the stated regularity assumptions. FITR shapes the trust region using local pullback-Fisher weights at the current center and replaces the lengthscales-based scaling in TuRBO by a diagonal approximation to the regularized pullback Fisher tensor. In our experiments, this diagonal FITR variant improves on TuRBO-REI on several GP benchmarks and remains competitive elsewhere. FITR does not introduce task-specific tuning and the same construction can be extended to other predictive families and to batch or multi-objective settings.

This perspective opens several directions for future work, including tighter boundary handling for full rotated Fisher regions and sharper theory for non-lengthscale kernels. The present work studies a theoretical aspect of BO and is intended for black-box optimization methodology; we do not foresee any broader societal impacts.

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A Gaussian Process

A Gaussian process (GP) is a distribution over functions, where we place a prior on the unknown objective and update our belief using the noisy observations [18].

We model the latent function as

$$f \sim \mathcal{GP}(m(\cdot), k(\cdot, \cdot')),$$

where $m(\cdot)$ is the prior mean and $k(\cdot, \cdot')$ is the covariance kernel. For observed input-output pairs $\mathcal{D}_t = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$, we use the Gaussian observation model

$$y_n = f(\mathbf{x}_n) + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma_\varepsilon^2).$$

The variance σ_ε^2 captures observation noise and appears as a diagonal correction to the kernel matrix.

Given $\mathcal{D}_t = \{(\mathbf{x}_n, y_n)\}_{n=1}^N$, the predictive distribution at a query point $\mathbf{x}$ is Gaussian

$$p(y \mid \mathbf{x}, \mathcal{D}_t) = \mathcal{N}(\mu_t(\mathbf{x}), \Sigma_t(\mathbf{x})),$$

where the posterior mean and variance are given by standard Gaussian conditioning identities [18]

$$\mu_t(\mathbf{x}) = m(\mathbf{x}) + k(\mathbf{x}, \mathbf{X}) (k(\mathbf{X}, \mathbf{X}) + \sigma_\varepsilon^2 \mathbf{I})^{-1} (\mathbf{y} - m(\mathbf{X})), \quad (11)$$

$$\Sigma_t(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - k(\mathbf{x}, \mathbf{X}) (k(\mathbf{X}, \mathbf{X}) + \sigma_\varepsilon^2 \mathbf{I})^{-1} k(\mathbf{X}, \mathbf{x}). \quad (12)$$

and the GP hyperparameters (for example, lengthscales in Λ and the noise level σ_ε^2) are estimated by maximizing the log marginal likelihood [18]

$$\log p(\mathbf{y} \mid \mathbf{X}, \Lambda) = -\frac{1}{2} (\mathbf{y} - m(\mathbf{X}))^\top K_\Lambda^{-1} (\mathbf{y} - m(\mathbf{X})) - \frac{1}{2} \log |K_\Lambda| - \frac{N}{2} \log(2\pi). \quad (13)$$

where $K_\Lambda = k(\mathbf{X}, \mathbf{X}) + \sigma_\varepsilon^2 \mathbf{I}$ [18].

B Common Acquisition Functions: Analytic and MC Forms

Section 2 introduces AF and its reparameterized form (2). When the acquisition is available in closed form, $\bar{\alpha}$ does not depend on ζ and (2) collapses to

$$\alpha(\mathbf{x}) = \mathbb{E}_\zeta[\bar{\alpha}(\boldsymbol{\theta}_{t,\mathbf{x}}; \boldsymbol{\xi})] = \bar{\alpha}(\boldsymbol{\theta}_{t,\mathbf{x}}; \boldsymbol{\xi}). \quad (14)$$

Table 1 shows some closed-form AF for a scalar Gaussian predictive, where $z = (\mu_{t,\mathbf{x}} - y^*)/\Sigma_{t,\mathbf{x}}^{1/2}$, $\Omega_{\mathcal{N}}$ is the Cumulative Distribution Function (CDF) and $\omega_{\mathcal{N}}$ is Probability Density Function (PDF) of normal distribution.

Table 1: Closed-form AF $\bar{\alpha}(\boldsymbol{\theta}_{t,\mathbf{x}}; \boldsymbol{\xi})$ under (14) for a scalar Gaussian predictive $\boldsymbol{\theta}_{t,\mathbf{x}} = (\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$. EI and PI use incumbent y^* ; UCB uses exploration weight $\kappa_{\text{UCB}} > 0$.

Acquisition	Closed-form $\bar{\alpha}(\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}}; \boldsymbol{\xi})$
EI	$(\mu_{t,\mathbf{x}} - y^*) \Omega_{\mathcal{N}}(z) + \Sigma_{t,\mathbf{x}}^{1/2} \omega_{\mathcal{N}}(z)$
PI	$\Omega_{\mathcal{N}}(z)$
UCB	$\mu_{t,\mathbf{x}} + \sqrt{\kappa_{\text{UCB}}} \Sigma_{t,\mathbf{x}}^{1/2}$

Following Wilson et al. [23], the Monte Carlo form of EI can be written as

$$\bar{\alpha}_{\text{EI}}(\boldsymbol{\theta}_{t,\mathbf{x}}, \zeta; \boldsymbol{\xi}) = \text{ReLU}(\mu_{t,\mathbf{x}} + \Sigma_{t,\mathbf{x}}^{1/2} \zeta - y^*). \quad (15)$$

For parallel BO [40] at batch input $\mathbf{X} = \{\mathbf{x}_1, \dots, \mathbf{x}_q\}$, we write the reparameterized sample as $\mathbf{y} = \boldsymbol{\mu}_{t,\mathbf{X}} + \mathbf{L}_{t,\mathbf{X}} \boldsymbol{\zeta}$, with $\mathbf{L}_{t,\mathbf{X}} = \text{chol}(\boldsymbol{\Sigma}_{t,\mathbf{X}})$ and $\boldsymbol{\zeta} \sim \mathcal{N}(0, \mathbf{I})$. Table 2 uses this batch notation because it matches the BoTorch implementation. In the scalar sequential setting of the main text, $q = 1$ and these expressions reduce to the usual one-point acquisitions.

For MC-UCB in the scalar case, the reparameterized form uses the identity $\mathbb{E}|Z| = \sqrt{2/\pi}$ for $Z \sim \mathcal{N}(0, 1)$, so $\sqrt{\kappa_{\text{UCB}} \pi/2} \Sigma_{t,\mathbf{x}}^{1/2} |Z|$ has expectation $\sqrt{\kappa_{\text{UCB}}} \Sigma_{t,\mathbf{x}}^{1/2}$.

Table 2: Utility a and integrand $\bar{\alpha}(\cdot, \zeta; \cdot)$ inside $\mathbb{E}_\zeta$ for reparameterized acquisitions under (2) for a Gaussian predictive [23, 40]. $\mathbf{1}^{+/-}$ denote the right/left-continuous Heaviside step functions; $\text{ReLU}(u) = \max(u, 0)$; $\text{sigmoid}(u) = (1 + e^{-u})^{-1}$; H is Shannon entropy. EI (Expected Improvement) and PI (Probability of Improvement) use incumbent threshold y^* ; PI uses temperature $\tau > 0$; the PI row is a sigmoid-smoothed surrogate for Monte Carlo gradients and is not the analytic PI functional $\Omega_{\mathcal{N}}(z)$. UCB uses exploration weight $\kappa_{\text{UCB}} > 0$. For ES (Entropy Search), subscripts a and b mark the query set and discretization points; $\mu_{b|a}$ and $L_{b|a}$ are the conditional posterior mean and Cholesky factor at b given an outcome at a . SR (Simple Regret) does not use extra parameters.

Acquisition	Utility $a(y; \theta_{t,\mathbf{x}}, \xi)$	Reparameterized form $\bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta; \xi)$
EI	$\max(\text{ReLU}(\mathbf{y} - y^*))$	$\max(\text{ReLU}(\mu_{t,\mathbf{x}} + L_{t,\mathbf{x}}\zeta - y^*))$
PI	$\max(\mathbf{1}^-(\mathbf{y} - y^*))$	$\max\left(\text{sigmoid}\left(\frac{\mu_{t,\mathbf{x}} + L_{t,\mathbf{x}}\zeta - y^*}{\tau}\right)\right)$
UCB	$\max(\mu_{t,\mathbf{x}} + \sqrt{\kappa_{\text{UCB}}\pi/2} \mathbf{y} - \mu_{t,\mathbf{x}})$	$\max(\mu_{t,\mathbf{x}} + \sqrt{\kappa_{\text{UCB}}\pi/2} L_{t,\mathbf{x}}\zeta)$
SR	$\max(\mathbf{y})$	$\max(\mu_{t,\mathbf{x}} + L_{t,\mathbf{x}}\zeta)$
ES	$-H(\mathbb{E}_{\mathbf{y}_b y_a}[\mathbf{1}^+(\mathbf{y}_b - \max(\mathbf{y}_b))])$	$-H\left(\mathbb{E}_{\zeta_b}\left[\text{softmax}\left(\frac{\mu_{b a} + L_{b a}\zeta_b}{\tau}\right)\right]\right)$

Algorithm 2: TuRBO TR weights calculation

Input: Center $\mathbf{x}_{t,c}$, data $\mathcal{D}_t$

Output: TR length weights $\bar{w}_t$, trained surrogate GP $\mathcal{M}_t$

Fit GP on $\mathcal{D}_t$ and extract ARD lengthscales $\lambda_{t,j}$ for all $j = 1, \dots, D_x$;

$\bar{w}_{t,j} \leftarrow \lambda_{t,j}$ for each $j = 1, \dots, D_x$;

return $\bar{w}_t = (\bar{w}_{t,j})_{j=1,\dots,D_x}$ and $\mathcal{M}_t$;

B.1 Trust-Region Bayesian Optimization (TuRBO)

TuRBO replaces a single global search step with local BO updates inside the trust region [11]. At each iteration t , a trust region (TR), $\mathcal{X}_t^{\text{TR}} \subseteq \mathcal{X}$, is defined by its center $\mathbf{x}_{t,c}$ and length κ_t . The TR is an axis-aligned box for which the dimension-wise length is calculated using the GP surrogate $\mathcal{M}_t$ with Automatic Relevance Determination (ARD) kernel. The next candidate is selected by maximizing an AF with bounds as

$$\mathbf{x}_{t+1} = \arg \max_{\mathbf{x} \in \mathcal{X}_t^{\text{TR}}} \alpha(\mathbf{x}; \mathcal{D}_t).$$

The TR length κ_t grows or shrinks using two counters. τ_{succ} tracks the number of consecutive improvements over the incumbent y^* , and τ_{fail} tracks the number of consecutive failures. If τ_{succ} reaches the threshold τ_{succ} , the region expands; if τ_{fail} reaches τ_{fail} , it contracts; both counters reset on any length change as

$$\kappa_{t+1} = \begin{cases} \min(2\kappa_t, \kappa_{\max}), & \text{if success count reaches } \tau_{\text{succ}}, \\ \frac{1}{2}\kappa_t, & \text{if failure count reaches } \tau_{\text{fail}}, \\ \kappa_t, & \text{otherwise.} \end{cases}$$

When $\kappa_t < \kappa_{\min}$ at any iteration, TR is said to have collapsed. It is then restarted around a new center location. Namura and Takemori [41] provide a Regional Expected Improvement (REI) acquisition function that can be used to select the next TR center location when the TR collapses. Eriksson et al. [11] also present a multi-region variant with multiple trust-region centers; however, we omit it here.

Algorithm 3: Trust-region length control and next candidate selection

Input: TR-weight routine `compute_tr_weights`, budget T , thresholds $\tau_{\text{succ}}^{\max}, \tau_{\text{fail}}^{\max}$, bounds

$\kappa_{\min}, \kappa_{\max}$ and initial length κ_{init}

Output: Best input $\mathbf{x}^*$ and value y^*

Initialize $\mathbf{x}_{0,c}, \kappa_0 \leftarrow \kappa_{\text{init}}, \tau_{\text{succ}} \leftarrow 0, \tau_{\text{fail}} \leftarrow 0, \mathcal{D}_0$, incumbent y^* ;

for $t = 0, \dots, T - 1$ **do**

 Get TR length weights $\bar{w}_j$ and $\mathcal{M}_t$ from `compute_tr_weights` with $(\mathbf{x}_{t,c}, \mathcal{D}_t)$;

Volume normalization;

$\bar{w}_j \leftarrow \bar{w}_j / (\prod_{k=1}^{D_x} \bar{w}_k)^{1/D_x}$ for each $j = 1, \dots, D_x$;

TR construction using upper and lower limits of $\mathcal{X}$;

for $j = 1, \dots, D_x$ **do**

$\alpha_j \leftarrow \max(x_j^c - \frac{\bar{w}_j \kappa_t}{2}, x_j^{\min})$;

$\beta_j \leftarrow \min(x_j^c + \frac{\bar{w}_j \kappa_t}{2}, x_j^{\max})$;

 Define $\mathcal{X}_t^{\text{TR}}$ by the dimension-wise limits $[\alpha_j, \beta_j]$ for all $j = 1, \dots, D_x$;

 Select next candidate $\mathbf{x}_{t+1} \leftarrow \arg \max_{\mathbf{x} \in \mathcal{X}_t^{\text{TR}}} \alpha(\mathbf{x}; \mathcal{D}_t)$;

 Evaluate $y_{t+1} = f_{\text{true}}(\mathbf{x}_{t+1})$ and update $\mathcal{D}_{t+1}$;

if $y_{t+1} > y^*$ **then**

$\tau_{\text{succ}} \leftarrow \tau_{\text{succ}} + 1; \tau_{\text{fail}} \leftarrow 0$;

 Update y^* ;

else

$\tau_{\text{succ}} \leftarrow 0; \tau_{\text{fail}} \leftarrow \tau_{\text{fail}} + 1$;

$\kappa_{t+1} \leftarrow \begin{cases} \min(2\kappa_t, \kappa_{\max}), & \tau_{\text{succ}} = \tau_{\text{succ}}^{\max} \\ \frac{1}{2}\kappa_t, & \tau_{\text{fail}} = \tau_{\text{fail}}^{\max} \\ \kappa_t, & \text{otherwise} \end{cases}$;

 Reset $\tau_{\text{succ}}, \tau_{\text{fail}} \leftarrow 0$ if κ changed;

if $\kappa_{t+1} < \kappa_{\min}$ **then**

 Restart around a new center; $\kappa_{t+1} \leftarrow \kappa_{\text{init}}$;

return $\mathbf{x}_{t+1}^*$ and y^* ;

C Implementation Details

For the experiments with a GP using an SE kernel, we use the Dimensionally Scaled LogNormal Priors (DSP) with mean $\sqrt{2} + \log(D_x) * 0.5$ and standard deviation $\sqrt{3}$. DSP [6] uses dimensionally scaled LogNormal priors on the lengthscales and optimizes AF over the full input domain $\mathcal{X}$, which is also the default implementation for SingleTaskGP in BoTorch v0.16.1 [15]. For the experiments with the GP using an IBNN kernel, we keep the BoTorch default depth 3.

TuRBO-REI combines TuRBO’s lengthscale-weighted trust region [11] with Regional Expected Improvement for choosing the trust-region center when the TR collapses [41]. Namura and Takemori [41] showed that REI improved TuRBO by selecting a new center in a region with high expected utility. BOUNCE [36] uses an adaptive subspace decomposition and can optimize mixed and combinatorial structures as well. FITR keeps the same trust-region scaling logic as TuRBO-REI (Algorithm 3) and also uses REI center selection at trust-region collapse. It replaces ARD lengthscale weights by the regularized diagonal pullback-Fisher weights from Algorithm 1. Section 5 makes clear that the main method in this paper is the diagonal axis-aligned approximation; we also test a full-tensor ablation (FullFITR). FullFITR uses whitening for Sobol sampling and a simple boundary-clipping strategy instead of polytope MCMC. Due to its additional computational cost, we evaluate FullFITR only on a subset of problems.

DSP, TuRBO-REI, FITR, and FullFITR are implemented in BoTorch [15] and use LogEI [1]. Inputs are normalized to $[0, 1]^{D_x}$, and outputs are standardized before fitting the GP. For each experiment, we start from 30 randomly sampled observed points. We use sequential evaluations ($q = 1$). The SE-kernel experiments are repeated over 11 independently seeded trials, whereas the IBNN experiments

are repeated over 5 independently seeded trials. AF is optimized using L-BFGS with 5 restarts and 256 raw samples. For FITR, the diagonal entries of the pullback Fisher tensor $g_t(\mathbf{x}_{t,c})$ are estimated with $S = 256$ MC samples using (9). These settings are fixed across tasks rather than tuned per benchmark.

The damping parameter is chosen automatically as $\varepsilon_t = D_x^{-1} \sum_{j=1}^{D_x} \hat{g}_{t,j} + 10^{-12}$. The resulting weights are volume-normalized before constructing the trust region. Using Algorithm 3, the trust-region length κ_t controls the search volume while the pullback tensor controls the local anisotropy. The trust-region length follows the standard TuRBO rule: it doubles after 10 consecutive improvements, halves after $\min\{\lceil \max(\frac{4}{q}, \frac{D_x}{q}) \rceil, 20\}$ consecutive failures, and restarts when the length drops below 0.5^7 . FullFITR uses the full regularized pullback Fisher tensor and optimizes LogEI in whitened coordinates. Candidate points are mapped back to the input cube and clipped at the boundary, which keeps the implementation compatible with standard BoTorch box optimization while exposing the additional cost of the full tensor. The experiments were run on an Nvidia GPU Tesla V100-SXM2-32GB.

C.1 Trust-region diagnostics

Figure 6 complements the objective-value plots with internal trust-region diagnostics. The trust-region length traces for FITR and TuRBO-REI remain qualitatively similar across problems, so the performance differences in Figure 4 are not explained by a different restart or length-adaptation schedule. For the diagonal FITR implementation, the logged anisotropy is the ratio between the largest and smallest volume-normalized coordinate weights used to shape the axis-aligned trust region, so values close to 1 indicate a nearly isotropic local region and larger values indicate stronger directional stretching. The diagnostics are shown using the experiment logs whose results are reported in Figure 4.

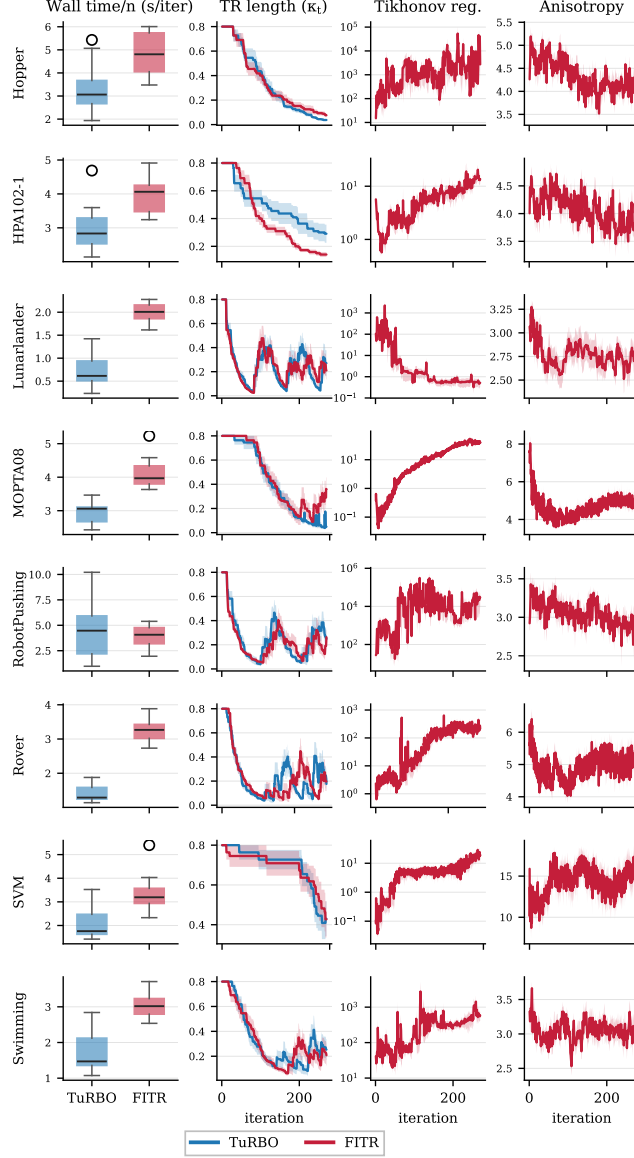


Figure 6: Per-problem diagnostics for TuRBO and FITR on the GP-SE benchmark suite. The first column reports the average time for getting the next candidate in BO iterations. The second column shows the trust-region length κ_t . The last two columns show the FITR-specific Tikhonov regularization and anisotropy diagnostics. Solid lines show means over the 11 seeded runs and shaded bands show standard error.

D Proofs

D.1 Interchange Conditions for Gaussian Reparameterizations

Differentiating under the expectation in Proposition 4.2 requires a valid interchange of gradient and expectation. We use standard reparameterization conditions from Wilson et al. [17]. The lemma below provides a sufficient condition for scalar Gaussian reparameterizations, where the AF integrand depends on the sampled output through the predictive distribution parameters θ . This is the case used later for MC-EI and the smoothed MC-PI surrogate.

Lemma D.1 (Generic Gaussian Reparameterization Condition). *Let Assumption 4.1 hold and let*

$$\bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta; \xi) = a(\mu_{t,\mathbf{x}} + \Sigma_{t,\mathbf{x}}^{1/2} \zeta; \xi)$$

be a scalar Gaussian reparameterized acquisition, with $\zeta \sim \mathcal{N}(0, I)$ independent of $\mathbf{x}$ [17, 40]. We assume

- (a) $\bar{\alpha}(\cdot, \zeta; \xi)$ is locally Lipschitz on $\mathcal{S}$ and differentiable almost everywhere for almost every ζ ;
- (b) $a(y; \xi)$ is globally Lipschitz in y .

Then $\mathbb{E}_\zeta [\|\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta; \xi)\|^2] < \infty$ for all $\theta_{t,\mathbf{x}} \in \mathcal{S}$.

Proof. Assumption (a) is the pathwise differentiability condition used in reparameterized acquisition optimization: the integrand admits gradients almost everywhere, which is sufficient for piecewise-smooth utilities such as MC-EI. For square-integrability, let L_a be the Lipschitz constant of a from assumption (b). The chain rule gives, for almost every ζ ,

$$\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha} = \nabla_y a(\mu_{t,\mathbf{x}} + \Sigma_{t,\mathbf{x}}^{1/2} \zeta; \xi) \cdot \nabla_{\theta_{t,\mathbf{x}}} (\mu_{t,\mathbf{x}} + \Sigma_{t,\mathbf{x}}^{1/2} \zeta). \quad (16)$$

Since a is L_a -Lipschitz in y , the first factor satisfies $\|\nabla_y a\| \leq L_a$. The second factor is affine in ζ , so $\|\nabla_{\theta_{t,\mathbf{x}}} (\mu_{t,\mathbf{x}} + \Sigma_{t,\mathbf{x}}^{1/2} \zeta)\| \leq A + B\|\zeta\|$ for finite constants A, B at a fixed $\theta_{t,\mathbf{x}}$. $\text{chol}(\cdot)$ is smooth on the positive-definite cone, since $\Sigma_{t,\mathbf{x}} \succ 0$ by Assumption 4.1(ii) [42]. Therefore $\|\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}\|^2 \leq L_a^2 (A + B\|\zeta\|)^2$, and since $\zeta \sim \mathcal{N}(0, I)$ has finite second moment the expectation is finite. $\square$

Remark (Necessity of $\Sigma \succ 0$). If the predictive variance degenerates, derivatives of the square-root factor can blow up, and the bound above would not hold. In the scalar case, $\nabla \Sigma^{1/2} = (\nabla \Sigma) / (2\Sigma^{1/2})$ is singular as $\Sigma \rightarrow 0$. A strictly positive noise level rules out this singularity for GP regression. Similar reasoning applies to the batch BO setting; however, we limit our discussion to the scalar case for simplicity in this work.

Remark (LogEI in the experimental regime). The experiments use the numerically stabilized LogEI implementation from BoTorch [1]. In the scalar Gaussian setting with strictly positive predictive variance, this implementation is smooth in the predictive parameters on any compact subset of $\mathbb{R} \times (0, \infty)$. The local interchange argument above therefore applies in the positive-variance regime considered in the experiments. We use EI for the closed-form sensitivity calculations in Appendix D.3, while LogEI is used in the implementation for numerical stability.

D.2 Proof of Proposition 4.2 (Acquisition Gradient Bound)

Let $\zeta \sim \mathcal{N}(0, I)$ be the independent base samples in the reparameterized acquisition from Lemma D.1. $\alpha(\mathbf{x}) = \mathbb{E}_\zeta [\bar{\alpha}(\theta_{t,\mathbf{x}}, \zeta; \xi)]$, where $\theta_{t,\mathbf{x}} = (\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$ are the GP posterior mean and variance at $\mathbf{x}$. $J_t(\mathbf{x}) = \partial \theta_{t,\mathbf{x}} / \partial \mathbf{x} \in \mathbb{R}^{D_\theta \times D_x}$ is the Jacobian of the posterior map.

Step 1 (Chain rule and interchange). By the assumed interchange of gradient and expectation in Proposition 4.2, justified for scalar Gaussian reparameterizations by Lemma D.1, the reparameterized acquisition satisfies

$$\nabla_{\mathbf{x}} \alpha(\mathbf{x}) = J_t(\mathbf{x})^\top \mathbb{E}_\zeta [\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}(\varphi_t(\mathbf{x}), \zeta; \xi)], \quad (17)$$

which also aligns with the discussion around Equation 5 of Section 3.2 in Wilson et al. [17]. For ease of notation, we write $\mathbb{E}_\zeta [\nabla_{\theta_{t,\mathbf{x}}} \bar{\alpha}(\varphi_t(\mathbf{x}), \zeta; \xi)]$ as $\mathbb{E}_\zeta [\nabla_{\theta} \bar{\alpha}(\cdot, \zeta)]$ in this proof, where the argument $\cdot$ denotes $\varphi_t(\mathbf{x})$.

Step 2 (Fisher metric insertion). Using Assumption 4.1(ii), let $\mathbf{g}_S(\theta_{t,\mathbf{x}}) := \mathcal{I}_S(\varphi_t(\mathbf{x})) \succ 0$. Using the identity $I = \mathbf{g}_S(\theta_{t,\mathbf{x}})^{1/2} \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1/2}$ from Amari and Nagaoka [24], we can decompose the gradient into a geometry-dependent factor A and an acquisition-dependent factor u :

$$\nabla_{\mathbf{x}} \alpha(\mathbf{x}) = \underbrace{(\mathbf{g}_S(\theta_{t,\mathbf{x}})^{1/2} J_t(\mathbf{x}))^\top}_{=: A^\top} \underbrace{\mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1/2} \mathbb{E}_\zeta [\nabla_{\theta} \bar{\alpha}(\cdot, \zeta)]}_{=: u}, \quad (18)$$

Step 3 (Cauchy–Schwarz). For any matrix $A \in \mathbb{R}^{p \times q}$ and vector $u \in \mathbb{R}^p$, $\|A^\top u\|_2^2 \leq \|A\|_F^2 \|u\|^2$. Applying it to the decomposition from Step 2 gives

$$\|\nabla_{\mathbf{x}} \alpha(\mathbf{x})\|_2 \leq \sqrt{\underbrace{\left\| \mathbf{g}_S(\theta_{t,\mathbf{x}})^{1/2} J_t(\mathbf{x}) \right\|_F^2}_{=\text{tr}(\mathbf{g}_t(\mathbf{x}))} \cdot \left\| \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1/2} \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)] \right\|^2}, \quad (19)$$

where using (5) we have $\text{tr}(\mathbf{g}_t(\mathbf{x})) = \text{tr}(J_t^\top \mathcal{I}_S(\varphi_t(\mathbf{x})) J_t) = \left\| \mathcal{I}_S^{1/2}(\varphi_t(\mathbf{x})) J_t \right\|_F^2$. And

$$\left\| \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1/2} \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)] \right\|^2 = \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)]^\top \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1} \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)]. \quad (20)$$

Step 4 (Jensen’s inequality). Considering $v = \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)]$, (20) can be rewritten as $f : v \mapsto v^\top \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1} v$, where f is convex for $\mathbf{g}_S(\theta_{t,\mathbf{x}}) \succ 0$. Using Jensen’s inequality we can write $f(\mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)]) \leq \mathbb{E}_\zeta[f(\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta))]$. Hence, we have

$$\left\| \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1/2} \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)] \right\|^2 \leq \mathbb{E}_\zeta[\nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)^\top \mathbf{g}_S(\theta_{t,\mathbf{x}})^{-1} \nabla_{\boldsymbol{\theta}} \bar{\alpha}(\cdot, \zeta)] = C_\alpha(\varphi_t(\mathbf{x})). \quad (21)$$

Combining (19) and (21) gives (6).

D.3 Gaussian Predictive Distribution

Corollary D.2 (Pullback FIM for a Gaussian Predictive Distribution). *Let Assumption 4.1 hold and suppose the predictive distribution is univariate Gaussian, $p(y|\mathbf{x}, \mathcal{D}_t) = \mathcal{N}(\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$, where $\mu_{t,\mathbf{x}} \in \mathbb{R}$ is the posterior mean and $\Sigma_{t,\mathbf{x}} > 0$ is the posterior variance, parameterized by $\boldsymbol{\theta}_{t,\mathbf{x}} = (\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}}) \in \mathcal{S} = \mathbb{R} \times \mathbb{R}_+$. The Fisher–Rao metric at this point on the predictive distribution manifold is*

$$\mathcal{I}_S(\boldsymbol{\theta}_{t,\mathbf{x}}) = \text{diag}(\Sigma_{t,\mathbf{x}}^{-1}, \frac{1}{2} \Sigma_{t,\mathbf{x}}^{-2}), \quad (22)$$

and the pullback tensor (5) reduces to

$$\mathbf{g}_t(\mathbf{x}) = \frac{\nabla \mu_{t,\mathbf{x}} \nabla \mu_{t,\mathbf{x}}^\top}{\Sigma_{t,\mathbf{x}}} + \frac{\nabla \Sigma_{t,\mathbf{x}} \nabla \Sigma_{t,\mathbf{x}}^\top}{2 \Sigma_{t,\mathbf{x}}^2}, \quad (23)$$

where ∇ denotes the gradient with respect to $\mathbf{x}$.

The first term, $\nabla \mu_{t,\mathbf{x}} \nabla \mu_{t,\mathbf{x}}^\top / \Sigma_{t,\mathbf{x}}$, is large where the posterior mean changes rapidly relative to uncertainty, so it captures the local exploitation geometry. The second term, $\nabla \Sigma_{t,\mathbf{x}} \nabla \Sigma_{t,\mathbf{x}}^\top / (2 \Sigma_{t,\mathbf{x}}^2)$, is large where the posterior variance changes rapidly, so it captures the local exploration geometry.

Proof of Corollary D.2. The log-likelihood of $\mathcal{N}(y; \mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$ is $\log p = -\frac{1}{2} \log(2\pi \Sigma_{t,\mathbf{x}}) - (y - \mu_{t,\mathbf{x}})^2 / (2 \Sigma_{t,\mathbf{x}})$. Differentiating with respect to the distribution parameters $\boldsymbol{\theta} = (\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$ gives the score functions

$$\frac{\partial \log p}{\partial \mu_{t,\mathbf{x}}} = \frac{y - \mu_{t,\mathbf{x}}}{\Sigma_{t,\mathbf{x}}}, \quad \frac{\partial \log p}{\partial \Sigma_{t,\mathbf{x}}} = \frac{(y - \mu_{t,\mathbf{x}})^2}{2 \Sigma_{t,\mathbf{x}}^2} - \frac{1}{2 \Sigma_{t,\mathbf{x}}}.$$

The secondary diagonal terms would be zero [13], and the FIM is a diagonal matrix

$$\mathcal{I}_S(\boldsymbol{\theta}_{t,\mathbf{x}}) = \text{diag}(\Sigma_{t,\mathbf{x}}^{-1}, \frac{1}{2} \Sigma_{t,\mathbf{x}}^{-2}).$$

The posterior map φ_t has Jacobian $J_t(\mathbf{x}) = [\nabla \mu_{t,\mathbf{x}}, \nabla \Sigma_{t,\mathbf{x}}]^\top \in \mathbb{R}^{2 \times D_x}$, where each row is the gradient of one parameter with respect to $\mathbf{x}$. Substituting into (5):

$$\begin{aligned} \mathbf{g}_t(\mathbf{x}) &= J_t^\top \text{diag}(\Sigma_{t,\mathbf{x}}^{-1}, \frac{1}{2} \Sigma_{t,\mathbf{x}}^{-2}) J_t \\ &= \Sigma_{t,\mathbf{x}}^{-1} \nabla \mu_{t,\mathbf{x}} \nabla \mu_{t,\mathbf{x}}^\top + \frac{1}{2} \Sigma_{t,\mathbf{x}}^{-2} \nabla \Sigma_{t,\mathbf{x}} \nabla \Sigma_{t,\mathbf{x}}^\top. \end{aligned}$$

□

D.4 Acquisition Sensitivity under a Scalar Gaussian Predictive

Substituting $\mathcal{I}_S^{-1} = \text{diag}(\Sigma_{t,\mathbf{x}}, 2\Sigma_{t,\mathbf{x}}^2)$ from (22) into the definition of C_α in (21) gives

$$C_\alpha(\boldsymbol{\theta}_t) = \mathbb{E}_\zeta \left[\Sigma_{t,\mathbf{x}} \left(\frac{\partial \bar{\alpha}}{\partial \mu_{t,\mathbf{x}}} \right)^2 + 2\Sigma_{t,\mathbf{x}}^2 \left(\frac{\partial \bar{\alpha}}{\partial \Sigma_{t,\mathbf{x}}} \right)^2 \right]. \quad (24)$$

For both analytic and Monte Carlo AFs, boundedness of C_α reduces to controlling partial derivatives of $\bar{\alpha}$ with respect to $(\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$.

Analytic acquisition functions. Let $z_t = (\mu_{t,\mathbf{x}} - y^*)/\sqrt{\Sigma_{t,\mathbf{x}}}$, Ω and ω be the cumulative distribution function (CDF) and probability density function (PDF) of the standard normal distribution.

For UCB [16, 43], $\bar{\alpha} = \mu_{t,\mathbf{x}} + \sqrt{\kappa \Sigma_{t,\mathbf{x}}}$ gives

$\partial \bar{\alpha} / \partial \mu_{t,\mathbf{x}} = 1$ and $\partial \bar{\alpha} / \partial \Sigma_{t,\mathbf{x}} = \sqrt{\kappa} / (2\sqrt{\Sigma_{t,\mathbf{x}}})$, so

$$C_\alpha^{\text{UCB}} = \Sigma_{t,\mathbf{x}} \left(1 + \frac{\kappa}{2} \right). \quad (25)$$

For EI [22, 44], $\bar{\alpha}^{\text{EI}} = (\mu_{t,\mathbf{x}} - y^*)\Omega(z_t) + \sqrt{\Sigma_{t,\mathbf{x}}}\omega(z_t)$ gives

$\partial z_t / \partial \Sigma_{t,\mathbf{x}} = -z_t / (2\Sigma_{t,\mathbf{x}})$, $\partial \bar{\alpha}^{\text{EI}} / \partial \mu_{t,\mathbf{x}} = \Omega(z_t)$ and $\partial \bar{\alpha}^{\text{EI}} / \partial \Sigma_{t,\mathbf{x}} = \omega(z_t) / (2\sqrt{\Sigma_{t,\mathbf{x}}})$. Using $\Omega \leq 1$ and $\omega \leq 1/\sqrt{2\pi}$,

$$C_\alpha^{\text{EI}} = \Sigma_{t,\mathbf{x}} \left(\Omega(z_t)^2 + \frac{1}{2} \omega(z_t)^2 \right) \leq \Sigma_{t,\mathbf{x}} \left(1 + \frac{1}{4\pi} \right), \quad (26)$$

For PI [22], $\bar{\alpha}^{\text{PI}} = \Omega(z_t)$ gives

$\partial \bar{\alpha}^{\text{PI}} / \partial \mu_{t,\mathbf{x}} = \omega(z_t) / \sqrt{\Sigma_{t,\mathbf{x}}}$ and $\partial \bar{\alpha}^{\text{PI}} / \partial \Sigma_{t,\mathbf{x}} = -z_t \omega(z_t) / (2\Sigma_{t,\mathbf{x}})$. Since $e^{-z^2} (1 + z^2/2) \leq 1$ for all $z \in \mathbb{R}$,

$$C_\alpha^{\text{PI}} = \omega(z_t)^2 \left(1 + \frac{z_t^2}{2} \right) \leq \frac{1}{2\pi}, \quad (27)$$

For bounded $\Sigma_{t,\mathbf{x}}$, the AF sensitivities in (25), (26), and (27) are bounded.

Remark on LogEI. The experiments use LogEI rather than analytic EI. In the strictly positive-variance regime enforced by Assumption 4.1, the stabilized LogEI implementation is smooth in the predictive parameters on compact subsets of $\mathbb{R} \times (0, \infty)$; see Remark D.1. The same local regularity argument therefore applies in the experimental setting, even though the closed-form sensitivity calculations below are written for EI because the formulas are shorter.

Lemma D.3 (Monte Carlo sensitivity bound). *Let Assumption 4.1 hold. Suppose $\bar{\alpha}(\boldsymbol{\theta}_t, \zeta; \boldsymbol{\xi})$ satisfies, for some $L_{\bar{\alpha}} > 0$ and almost every ζ ,*

$$\left| \frac{\partial \bar{\alpha}}{\partial \mu_{t,\mathbf{x}}} \right| \leq L_{\bar{\alpha}}, \quad \left| \frac{\partial \bar{\alpha}}{\partial \Sigma_{t,\mathbf{x}}} \right| \leq \frac{L_{\bar{\alpha}} |\zeta|}{2\sqrt{\Sigma_{t,\mathbf{x}}}}. \quad (28)$$

Then

$$C_\alpha(\boldsymbol{\theta}_t) \leq \frac{3}{2} L_{\bar{\alpha}}^2 \Sigma_{t,\mathbf{x}}. \quad (29)$$

Proof. Substituting (28) into (24) and using $\mathbb{E}[\zeta^2] = 1$,

$$C_\alpha \leq \mathbb{E}_\zeta \left[L_{\bar{\alpha}}^2 \Sigma_{t,\mathbf{x}} + \frac{1}{2} L_{\bar{\alpha}}^2 \Sigma_{t,\mathbf{x}} \zeta^2 \right] = \frac{3}{2} L_{\bar{\alpha}}^2 \Sigma_{t,\mathbf{x}}. \quad \square$$

For MC-EI [17, 23], $\bar{\alpha}^{\text{EI}} = \text{ReLU}(\mu_{t,\mathbf{x}} + \sqrt{\Sigma_{t,\mathbf{x}}} \zeta - y^*)$ from Table 2 gives

$|\partial \bar{\alpha}^{\text{EI}} / \partial \mu_{t,\mathbf{x}}| = \mathbf{1}^+ \leq 1$ and $|\partial \bar{\alpha}^{\text{EI}} / \partial \Sigma_{t,\mathbf{x}}| = \mathbf{1}^+ |\zeta| / (2\sqrt{\Sigma_{t,\mathbf{x}}}) \leq |\zeta| / (2\sqrt{\Sigma_{t,\mathbf{x}}})$, so (28) holds with $L_{\bar{\alpha}} = 1$ and $C_\alpha^{\text{MC-EI}} \leq \frac{3}{2} \Sigma_{t,\mathbf{x}}$.

For MC-PI [17], $\bar{\alpha}^{\text{PI}} = \text{sigmoid}((\mu_{t,\mathbf{x}} + \sqrt{\Sigma_{t,\mathbf{x}}} \zeta - y^*)/\tau)$

and both partial derivatives have the term $\text{sigmoid}'(\cdot)/\tau \leq 1/(4\tau)$, so $L_{\bar{\alpha}} = 1/(4\tau)$. Using $\mathbb{E}[\zeta^2] = 1$, we get $C_\alpha^{\text{MC-PI}} \leq 3\Sigma_{t,\mathbf{x}}/(32\tau^2)$.

For MC-UCB [17], $\bar{\alpha}^{\text{UCB}} = \mu_{t,\mathbf{x}} + \sqrt{\kappa\pi/2} \sqrt{\Sigma_{t,\mathbf{x}}} |\zeta|$ from Table 2,

which depends on $\mu_{t,\mathbf{x}}$ outside the reparameterized sample, so the second condition in (28) cannot hold with the same $L_{\bar{\alpha}}$ as the first. Direct pathwise differentiation gives $\partial\bar{\alpha}^{\text{UCB}}/\partial\mu_{t,\mathbf{x}} = 1$ and $\partial\bar{\alpha}^{\text{UCB}}/\partial\Sigma_{t,\mathbf{x}} = \sqrt{\kappa\pi/2} |\zeta|/(2\sqrt{\Sigma_{t,\mathbf{x}}})$; substituting into (24) and using $\mathbb{E}[\zeta^2] = 1$, we get $C_{\alpha}^{\text{MC-UCB}} = \Sigma_{t,\mathbf{x}}(1 + \frac{\kappa\pi}{4})$.

Table 3: Calculated bounds on $C_{\alpha}(\theta_{t,\mathbf{x}})$ for the scalar Gaussian predictive. If $\Sigma_{t,\mathbf{x}}$ is bounded, which holds for any GP with finite prior variance, the shown AF sensitivities are also bounded. Analytic PI is bounded globally on $\mathcal{S}$. Since all entries depend only on $(\mu_{t,\mathbf{x}}, \Sigma_{t,\mathbf{x}})$, they apply to any surrogate with a Gaussian predictive.

Acquisition	Analytic bound on C_{α}	Monte Carlo bound on C_{α}
EI	$\Sigma_{t,\mathbf{x}}(1 + 1/(4\pi))$	$\frac{3}{2}\Sigma_{t,\mathbf{x}}$
UCB	$\Sigma_{t,\mathbf{x}}(1 + \kappa/2)$	$\Sigma_{t,\mathbf{x}}(1 + \kappa\pi/4)$
PI	$1/(2\pi)$	$\frac{3}{32\pi^2}\Sigma_{t,\mathbf{x}}$

Remark (Extension to multivariate outputs and parallel BO). *For a multivariate Gaussian predictive distribution $\mathcal{N}(\mu_t, \Sigma_t)$ with $\mu_t \in \mathbb{R}^{D_y}$ and $\Sigma_t \in \mathbb{R}^{D_y \times D_y}$ (as in Multi-Objective Bayesian Optimization (MOBO) with D_y correlated outputs), the Fisher–Rao metric takes the block-diagonal form $\mathcal{I}_{\mathcal{S}} = \text{diag}(\Sigma_t^{-1}, \frac{1}{2}\Sigma_t^{-1} \otimes \Sigma_t^{-1})$ [13], and (23) extends to*

$$\mathbf{g}_t(\mathbf{X}) = J_{\mu}^{\top} \Sigma_t^{-1} J_{\mu} + \frac{1}{2} J_{\Sigma}^{\top} (\Sigma_t^{-1} \otimes \Sigma_t^{-1}) J_{\Sigma}, \quad (30)$$

where J_{μ} and J_{Σ} are the Jacobians of the vectorized mean and covariance with respect to $\mathbf{X}$.

We limit our discussion to the scalar case for simplicity. However, following Wang et al. [40] or Balandat et al. [15], one can apply a similar analysis to obtain a bound for the acquisition sensitivity $C_{\alpha}^{(q)}(\theta_{t,\mathbf{x}})$ for Multi-Objective Bayesian Optimization or parallel BO with $q > 1$.

D.5 Pullback FIM Trace Under the Critical Lengthscale

Distance concentration. We assume that the initial training points $\mathbf{x}_i$ for $i = 1, \dots, N$ are drawn i.i.d. from Uniform $[0, 1]^{D_x}$. Let $d := \|\mathbf{x} - \mathbf{x}_n\|$. Since $\mathbb{E}[(x_j - \frac{1}{2})^2] = (x_j - \frac{1}{2})^2 + \frac{1}{12}$ for $x_{n,j} \sim \text{Uniform}[0, 1]$, we have

$$\frac{d}{\sqrt{D_x}} \xrightarrow{\text{a.s.}} \sqrt{v_{\mathbf{x}}}, \quad v_{\mathbf{x}} := \frac{1}{D_x} \sum_{j=1}^{D_x} [(x_j - \frac{1}{2})^2 + \frac{1}{12}] \in [\frac{1}{12}, \frac{1}{3}].$$

Hence $d = \Theta(\sqrt{D_x})$ almost surely. Here $f = \Theta(g)$ means there exist $c_1, c_2 > 0$ and d_0 such that $c_1 g(d) \leq f(d) \leq c_2 g(d)$ for all $d \geq d_0$.

FIM trace for the SE kernel. For a GP with SE kernel $k_{\lambda}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\|\mathbf{x} - \mathbf{x}'\|^2/(2\lambda^2))$, let $K_{nn'} = k_{\lambda}(\mathbf{x}_n, \mathbf{x}_{n'})$ and $\alpha = (K + \sigma_{\epsilon}^2 I)^{-1} \mathbf{y}$. Then

$$\nabla_{\mathbf{x}} \mu_t(\mathbf{x}) = \sum_{n=1}^N \alpha_n \nabla_{\mathbf{x}} k_{\lambda}(\mathbf{x}, \mathbf{x}_n), \quad \nabla_{\mathbf{x}} k_{\lambda}(\mathbf{x}, \mathbf{x}_n) = -\frac{k_{\lambda}(\mathbf{x}, \mathbf{x}_n)}{\lambda^2} (\mathbf{x} - \mathbf{x}_n).$$

By Corollary D.2,

$$\text{tr}(\mathbf{g}_t) = \frac{\|\nabla \mu_t\|^2}{\Sigma_t} + \frac{\|\nabla \Sigma_t\|^2}{2\Sigma_t^2},$$

and both terms scale with k_{λ}/λ^2 . Substituting $\lambda = cD_x^p$ and $d = \Theta(\sqrt{D_x})$ gives

$$k_{\lambda}(\mathbf{x}, \mathbf{x}_n) \xrightarrow{\text{a.s.}} \sigma_f^2 \exp\left(-\frac{v_{\mathbf{x}}}{2c^2} D_x^{1-2p}\right).$$

This results in three cases:

Case $p < 1/2$. The exponent $v_{\mathbf{x}} D_x^{1-2p}/(2c^2) \rightarrow \infty$, so $k_{\lambda} \rightarrow 0$ exponentially. Hence $\nabla_{\mathbf{x}} k_{\lambda} \rightarrow 0$ exponentially, and therefore $\nabla \mu_t \rightarrow 0$ and $\text{tr}(\mathbf{g}_t) \rightarrow 0$ exponentially.

Case $p > 1/2$. The exponent $\rightarrow 0$, so $k_{\lambda} \rightarrow \sigma_f^2$ (constant). The gradient $\nabla_{\mathbf{x}} k_{\lambda} \propto k_{\lambda} \cdot (\mathbf{x} - \mathbf{x}_n)/\lambda^2$. Since each coordinate difference is $\Theta(1)$ while $\lambda^2 = c^2 D_x^{2p}$, each entry scales as $\Theta(D_x^{-2p})$ and the full gradient norm scales as $\Theta(D_x^{1/2-2p}) \rightarrow 0$. Hence $\nabla \mu_t \rightarrow 0$ and $\text{tr}(\mathbf{g}_t) \rightarrow 0$ at a polynomial rate.

Case $p = 1/2$. The exponent converges to $v_{\mathbf{x}}/(2c^2) \in (0, \infty)$, so $k_{\lambda} \rightarrow \gamma_c := \sigma_f^2 e^{-v_{\mathbf{x}}/(2c^2)} \in (0, \sigma_f^2)$. The Gram matrix concentrates to $K_{\infty} = (\sigma_f^2 + \sigma_{\varepsilon}^2 - \bar{\gamma})I + \bar{\gamma} \mathbf{1}\mathbf{1}^{\top}$, where $\bar{\gamma} = \sigma_f^2 e^{-1/(12c^2)}$ is the pairwise limit for i.i.d. training pairs. Since $\lambda_{\min}(K_{\infty}) > 0$, we have $\alpha = \Theta(1)$. Each entry of $\nabla_{\mathbf{x}} k_{\lambda}$ scales as $\Theta(D_x^{-1})$, so the full gradient norm scales as $\Theta(D_x^{-1/2})$ and summing D_x squared terms gives $\text{tr}(\mathbf{g}_t) = \Theta(D_x^{-1})$.

In all three cases, the trace of $\mathbf{g}_t$ is governed solely by whether k_{λ} is able to model meaningful correlation or not. $\lambda = \Theta(\sqrt{D_x})$ is the unique scaling for which $\text{tr}(\mathbf{g}_t)$ stays at the critical $\Theta(D_x^{-1})$ order instead of decaying faster.

FIM trace at $p = 1/2$ and acquisition gradient bound. With $\alpha = \Theta(1)$ and $\gamma_c = \Theta(1)$, substituting into Proposition 4.2 gives

$$\|\nabla_{\mathbf{x}} \alpha\|^2 \leq C_{\alpha} \cdot \Theta(D_x^{-1}).$$

Since C_{α} is uniformly bounded in $\mathbf{x}$ and D_x for standard acquisitions (Appendix D.3), the universal bound scales as $\Theta(D_x^{-1})$ rather than collapsing exponentially. For $p < 1/2$, the trace collapses exponentially, while for $p > 1/2$ it decays polynomially. In both off-critical regimes, the AF gradient norm still vanishes.

Matérn kernels. The same argument applies to isotropic Matérn- ν kernel, since it depends on inputs only through $d = \|\mathbf{x} - \mathbf{x}'\|/\lambda$ and the same concentration gives $d \rightarrow \sqrt{v_{\mathbf{x}} D_x}/\lambda$. Thus the non-degenerate scaling remains $\lambda = \Theta(\sqrt{D_x})$. The difference appears when $\lambda \ll \sqrt{D_x}$: using $k_{\nu}(d) \sim d^{\nu-1/2} e^{-\sqrt{2\nu}d}$ as $d \rightarrow \infty$, the trace decays as $\exp(-c_{\nu} \sqrt{D_x}/\lambda)$, whereas for the SE kernel it decays as $\exp(-c_1 D_x/\lambda^2)$. Since $D_x/\lambda^2 \gg \sqrt{D_x}/\lambda$ in that regime, the SE trace collapses strictly faster, matching the empirical and theoretical results documented by Xu et al. [7].

D.6 Connection between TuRBO and FITR

Consider a GP with SE kernel with Automatic Relevance Determination (ARD) $k_{\lambda}(\mathbf{x}, \mathbf{x}') = \sigma_f^2 \exp(-\sum_j (x_j - x'_j)^2/(2\lambda_j^2))$ evaluated at the trust-region center $\mathbf{x}_c$.

At a local optimum of μ_t , $\nabla \mu_t(\mathbf{x}_c) \approx 0$, so the mean-gradient term in the pullback FIM (Corollary D.2) vanishes. The diagonal of $\mathbf{g}_t(\mathbf{x}_c)$ is then dominated by the variance-gradient term:

$$[g_t(\mathbf{x}_c)]_{jj} \approx \frac{1}{2\Sigma_t(\mathbf{x}_c)^2} \left(\frac{\partial \Sigma_t}{\partial x_j} \right)^2.$$

For the SE kernel, $\partial k_{\lambda}(\mathbf{x}, \mathbf{x}')/\partial x_j = -(x_j - x'_j)k_{\lambda}(\mathbf{x}, \mathbf{x}')/\lambda_j^2$, so $\partial \Sigma_t/\partial x_j$ scales proportionally to λ_j^{-2} . Using the definition of the pullback tensor, the diagonal entry $[g_t(\mathbf{x}_c)]_{jj}$ then scales like λ_j^{-4} . The ellipsoid induced by the pullback tensor is therefore longer in dimensions with longer lengthscales and shorter in dimensions with shorter lengthscales:

$$\{(\mathbf{x} - \mathbf{x}_c)^{\top} \text{diag}(\mathbf{g}_t(\mathbf{x}_c))(\mathbf{x} - \mathbf{x}_c) \leq \delta^2\}.$$

The radius in dimension j is proportional to $[g_t]_{jj}^{-1/2}$. TuRBO's trust region $\{|x_j - x_{c,j}| \leq L\lambda_j\}$ is not an exact diagonal-FIM box, but it preserves the same qualitative ordering: directions with larger ARD lengthscales receive wider search lengths. This motivates extending TuRBO to a location-dependent trust region whose scaling is determined by predictive sensitivities rather than explicit kernel lengthscales.

D.7 Local KL Expansion for Pullback-Fisher Trust Regions

This subsection gives the local probabilistic interpretation of the trust-region ellipsoid from Section 5. The key point is that the full pullback Fisher tensor defines the second-order term of the KL divergence between nearby predictive posteriors, so a trust region based on this tensor limits local predictive change rather than raw Euclidean displacement.

Fix an iteration t and center $\mathbf{x}_{t,c} \in \mathcal{X}$. Let

$$\boldsymbol{\theta}_{t,c} := \varphi_t(\mathbf{x}_{t,c}), \quad \delta := \mathbf{x} - \mathbf{x}_{t,c}.$$

Consider the KL divergence between the predictive posterior at the center and the predictive posterior at a nearby input:

$$\mathcal{K}_t(\delta) := D_{\text{KL}}(p(y; \boldsymbol{\theta}_{t,c}) \parallel p(y; \varphi_t(\mathbf{x}_{t,c} + \delta))). \quad (31)$$

Step 1: KL expansion in parameter space. For a smooth parametric family $p(y; \boldsymbol{\theta})$, the KL divergence admits the standard second-order expansion around $\boldsymbol{\theta}_{t,c}$ [13, 24]:

$$D_{\text{KL}}(p(y; \boldsymbol{\theta}_{t,c}) \parallel p(y; \boldsymbol{\theta}_{t,c} + \Delta\boldsymbol{\theta})) = \frac{1}{2} \Delta\boldsymbol{\theta}^\top \mathcal{I}_S(\boldsymbol{\theta}_{t,c}) \Delta\boldsymbol{\theta} + o(\|\Delta\boldsymbol{\theta}\|^2). \quad (32)$$

Step 2: Pull back to input space. Since φ_t is C^1 by Assumption 4.1, we have the first-order expansion

$$\varphi_t(\mathbf{x}_{t,c} + \delta) = \boldsymbol{\theta}_{t,c} + J_t(\mathbf{x}_{t,c}) \delta + o(\|\delta\|). \quad (33)$$

Substituting

$$\Delta\boldsymbol{\theta} = J_t(\mathbf{x}_{t,c}) \delta + o(\|\delta\|)$$

into (32) gives

$$\begin{aligned} \mathcal{K}_t(\delta) &= \frac{1}{2} \delta^\top J_t(\mathbf{x}_{t,c})^\top \mathcal{I}_S(\boldsymbol{\theta}_{t,c}) J_t(\mathbf{x}_{t,c}) \delta + o(\|\delta\|^2) \\ &= \frac{1}{2} \delta^\top \mathbf{g}_t(\mathbf{x}_{t,c}) \delta + o(\|\delta\|^2), \end{aligned} \quad (34)$$

where the second line uses Definition 4.1.

Equation (34) shows that the ellipsoid

$$\delta^\top \mathbf{g}_t(\mathbf{x}_{t,c}) \delta \leq \kappa_t^2 \quad (35)$$

is, to second order, a local KL trust region around the predictive posterior at $\mathbf{x}_{t,c}$. This interpretation applies to the full pullback Fisher tensor. The FITR implementation in Section 5 uses a diagonal, regularized approximation of (35) to obtain an axis-aligned box that is computationally compatible with TuRBO-style trust-region updates.

D.8 Pullback FIM Diagonal via Backpropagation

The estimator (9) computes the diagonal of the pullback Fisher tensor by differentiating the log predictive likelihood with respect to the input $\mathbf{x}$ rather than the distribution parameters $\boldsymbol{\theta}$. The connection follows from the chain rule and the definition of the FIM.

By the chain rule, the partial derivative of the log predictive likelihood with respect to input coordinate x_j is

$$\frac{\partial}{\partial x_j} \log p(y; \varphi_t(\mathbf{x})) = (\nabla_{\boldsymbol{\theta}} \log p(y; \boldsymbol{\theta}))^\top \frac{\partial \varphi_t(\mathbf{x})}{\partial x_j}. \quad (36)$$

Squaring both sides and taking the expectation over $y \sim p(y; \varphi_t(\mathbf{x}))$,

$$\mathbb{E} \left[\left(\frac{\partial}{\partial x_j} \log p(y; \varphi_t(\mathbf{x})) \right)^2 \right] = \left(\frac{\partial \varphi_t}{\partial x_j} \right)^\top \underbrace{\mathbb{E} [\nabla_{\boldsymbol{\theta}} \log p \nabla_{\boldsymbol{\theta}} \log p^\top]}_{= \mathcal{I}_S(\varphi_t(\mathbf{x}))} \left(\frac{\partial \varphi_t}{\partial x_j} \right) = [\mathbf{g}_t(\mathbf{x})]_{jj}, \quad (37)$$

where the last equality uses the definition of the FIM (3) and the pullback tensor (5). The diagonal entry of the pullback Fisher tensor is therefore the expected squared partial derivative of the log predictive likelihood with respect to x_j , computed at the predictive posterior $\varphi_t(\mathbf{x})$.

Replacing the expectation by an average over S draws $\tilde{y}_s \sim p(y; \varphi_t(\mathbf{x}))$ and evaluating the partial derivatives by backpropagation through the log-likelihood gives the estimator (9). This is the outer-product-of-gradients form of the FIM, similar to [25], applied to the input coordinates $\mathbf{x}$ instead of the distribution parameters $\boldsymbol{\theta}$.

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